Numerical Methods for First-Order ODEs

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This chapter is about computing solutions of the **initial value problem** (IVP)

$$y' = f(x, y), \qquad y(x_0) = y_0, \qquad x_0 \le x \le b$$

when no formula for $y(x)$ is available. It starts from the simplest method (Euler) and builds up to the ideas used inside professional solvers: higher-order Runge–Kutta methods, error estimation, stability, stiffness, adaptive step size and multistep methods.

Prerequisite: First-Order Differential Equations, sections 1–5.

Chapter contents

	Section	What it covers
1	Euler's method	Three derivations, local and global error, a proof that Euler converges, examples where it fails
2	Second-order Runge–Kutta methods	Heun, midpoint and Ralston; deriving the whole two-stage family from the Taylor series
3	The classical Runge–Kutta method (RK4)	RK4, Butcher tableaux, order conditions, the Butcher barriers
4	Error, order and efficiency	Measuring order, accuracy per unit of work, Richardson extrapolation, round-off error
5	Stability and stiff equations	Amplification factors, stability regions, implicit methods, A- and L-stability, stiffness
6	Adaptive step size	Error estimators, embedded pairs (Heun–Euler, Bogacki–Shampine), step-size control

	Section	What it covers
7	Multistep methods	Adams–Bashforth, Adams–Moulton, predictor–corrector, zero-stability and Dahlquist's theorems
8	Systems and higher-order equations	Vector ODEs, oscillators, the pendulum, predator–prey models, conserved quantities

```
import numpy as np
import matplotlib.pyplot as plt

plt.rcParams["figure.figsize"] = (7, 4.5)
plt.rcParams["axes.grid"] = True
np.set_printoptions(precision=6, suppress=True)
```

1. Why numerical methods?

The analytical methods (separable, linear, exact equations) cover only a small fraction of the equations that occur in practice. Even an innocent-looking equation such as the Riccati equation

$$y' = x^2 + y^2$$

has no solution in terms of elementary functions. Other times a formula exists but is implicit, or contains integrals nobody can evaluate. A numerical method only needs to be able to **evaluate** $f(x, y)$ at given numbers, so it works for every equation.

The price is that the result is a table of approximate values, not a formula, and it contains **errors**. Most of this chapter is about understanding, estimating and controlling those errors.

2. When does the problem have a solution?

Before computing a solution it is worth knowing that there is exactly one solution to compute.

Lipschitz condition. f satisfies a Lipschitz condition in y on a region D if there is a constant L such that

$$|f(x, y_1) - f(x, y_2)| \le L\,|y_1 - y_2| \qquad \text{for all } (x, y_1), (x, y_2) \in D.$$

If $\partial f/\partial y$ exists and is bounded on D (and D is convex in y), the mean value theorem shows that $L = \max_D |\partial f/\partial y|$ works.

Theorem (Picard–Lindelöf). If f is continuous and satisfies a Lipschitz condition in y on a rectangle around (x_0, y_0), then the IVP has **exactly one** solution on some interval around x_0.

Equation	$\partial f/\partial y$	What happens
$y' = x + y$	1	$L = 1$ everywhere: one solution, defined for all x
$y' = y^2$, $y(0) = 1$	$2y$	Lipschitz only on bounded y-ranges: the solution $y = 1/(1 - x)$ exists only for $x < 1$ (it blows up)

Equation	$\partial f/\partial y$	What happens
$y' = \sqrt{\lvert y\rvert}$, $y(0) = 0$	$\frac{1}{2}\lvert y\rvert^{-1/2}$, unbounded at $y = 0$	**Not** Lipschitz: both $y = 0$ and $y = x^2/4$ ($x \geq 0$) are solutions

The last two rows are warnings. A numerical method applied to $y' = y^2$ happily produces finite numbers beyond $x = 1$, and applied to $y' = \sqrt{\lvert y\rvert}$ it returns $y_n = 0$ forever without telling you a second solution exists. **The theory tells you when the numbers can be trusted.**

3. How sensitive is the solution to its starting value?

Let y and z solve the same equation with slightly different starting values. Using the Lipschitz condition and Grönwall's inequality one can show

$$|y(x) - z(x)| \leq e^{L(x-x_0)}\,|y_0 - z_0|.$$

So a small change in the starting value can grow, at worst, exponentially with rate L. Whether it actually grows depends on the **sign** of $\partial f/\partial y$:

- $\partial f/\partial y > 0$: neighbouring solutions **spread apart**;
- $\partial f/\partial y < 0$: neighbouring solutions **draw together**.

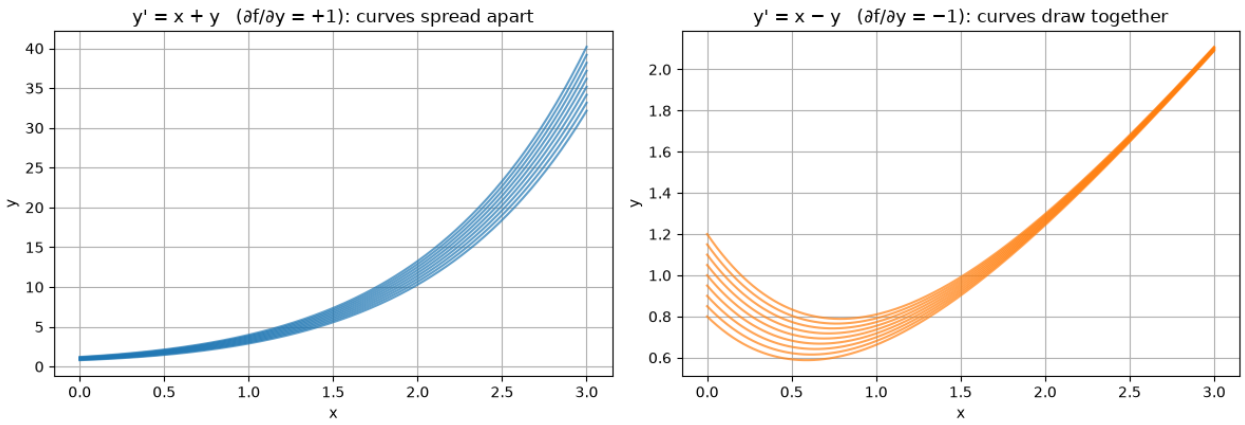
This matters for numerical methods because every step makes a small error, which moves the computed solution onto a *neighbouring* solution curve. If neighbouring curves spread apart, those errors are amplified as the computation proceeds; if they draw together, old errors die out.

The plots show solutions starting from $y_0 \in [0.8, 1.2]$ for $y' = x + y$ (solutions $y = Ce^x - x - 1$) and $y' = x - y$ (solutions $y = x - 1 + Ce^{-x}$).

```
x = np.linspace(0, 3, 200)
fig, axes = plt.subplots(1, 2, figsize=(12, 4.2))
for y0 in np.linspace(0.8, 1.2, 9):
    C = y0 + 1
    axes[0].plot(x, C * np.exp(x) - x - 1, color="C0", alpha=0.7)
    axes[1].plot(x, x - 1 + C * np.exp(-x), color="C1", alpha=0.7)
axes[0].set_title("y' = x + y   (f/y = +1): curves spread apart")
axes[1].set_title("y' = x - y   (f/y = -1): curves draw together")
for ax in axes:
    ax.set_xlabel("x"); ax.set_ylabel("y")
plt.tight_layout()

for y0 in (0.8, 1.2):
    C = y0 + 1
    print(f"y0 = {y0}:  y(3) = {C*np.exp(3) - 4:8.4f} for x + y,   {2 + C*np.exp(-3):.4f} for x - y")
```

```
y0 = 0.8:  y(3) =  32.1540 for x + y,   2.0896 for x - y
y0 = 1.2:  y(3) =  40.1882 for x + y,   2.1095 for x - y
```



A spread of 0.4 in the starting value becomes a spread of $0.4e^3 \approx 8$ at $x = 3$ for the first equation, but shrinks to $0.4e^{-3} \approx 0.02$ for the second.

4. Discretising the problem

A numerical method replaces the unknown function by its values on a grid.

Symbol	Meaning
h	step size
$x_n = x_0 + nh$	grid points, $n = 0, 1, \ldots, N$ with $N = (b - x_0)/h$
$y(x_n)$	the **exact** solution at x_n (unknown)
y_n	the **computed** approximation to $y(x_n)$
$f_n = f(x_n, y_n)$	the slope evaluated at the computed point
$e_n = y(x_n) - y_n$	the global error at x_n

5. Every method approximates an integral

Integrating $y' = f(x, y)$ over one step gives the **exact** relation

$$y(x_{n+1}) = y(x_n) + \int_{x_n}^{x_{n+1}} f(x, y(x))\, dx.$$

The integral cannot be computed exactly because $y(x)$ inside it is unknown. Each method is a different way of approximating it, and most correspond to a familiar quadrature rule:

Method	Approximates the integral by	Quadrature rule	Section
Euler	slope at the left end	left rectangle	1
Backward Euler	slope at the right end	right rectangle	5
Midpoint	slope at the middle	midpoint rule	2
Heun, trapezoidal	average of the end slopes	trapezoidal rule	2, 5
RK4	weighted average of four slopes	Simpson's rule	3
Adams methods	integral of a polynomial through earlier slopes	interpolatory quadrature	7

6. Classifying methods

- **One-step or multistep.** A one-step method (Euler, Runge–Kutta) computes y_{n+1} from (x_n, y_n) only. A multistep method also reuses $y_{n-1}, y_{n-2}, \ldots$ and their slopes.
- **Explicit or implicit.** An explicit method gives y_{n+1} by a formula. An implicit method gives an *equation* for y_{n+1} that has to be solved at every step. Implicit methods cost more per step but can be far more stable.
- **Fixed or adaptive step.** A fixed-step method uses the same h throughout. An adaptive method estimates its own error and changes h as it goes.

7. Three properties every method needs

- **Consistency.** The formula is a correct approximation of the ODE: the error made in a single step, divided by h, tends to 0 as $h \to 0$.
- **Stability.** Errors made in early steps are not amplified without bound by later steps.
- **Convergence.** The computed values approach the exact solution as the step shrinks: $\max_n |y(x_n) - y_n| \to 0$ as $h \to 0$.

The central theorem of the subject says, roughly,

$$\textbf{consistency} + \textbf{stability} \implies \textbf{convergence}.$$

A convergent method has **order** p if its global error behaves like Ch^p. Sections 1–4 are mostly about consistency and order, Sections 5 and 7 about stability.

8. The test problem

To be able to measure errors exactly, most sections use the same test problem:

$$y' = x + y, \qquad y(0) = 1, \qquad \text{exact solution } y = 2e^x - x - 1.$$